Optimizing Consumer Credit Risk Scoring: Leakage-Free Data Engineering, Calibrated Gradient Boosters, and Ensembled Probability of Default Estimation

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Abstract*—Accurate quantification of credit default risk is foundational to modern financial stability, capital allocation, and regulatory solvency under the Basel Accords. While contemporary machine learning literature frequently reports near-perfect classification performance on peer-to-peer lending benchmarks, such figures almost ubiquitously stem from subtle post-origination data leakage (e.g., training on post-issuance recoveries, payment trajectories, and debt settlement tags). In this paper, we propose a rigorous, leak-free, end-to-end machine learning credit risk modeling framework evaluated on 100,000 real-world consumer loans from the Lending Club portfolio. Our methodology integrates: (1) systematic auditing and removal of 40 post-origination leakage features; (2) a quantitative multi-criteria data quality framework improving portfolio health from 99.39% to 100.00%; (3) domain-driven financial interaction engineering capturing debt-burden velocity, delinquency severity, and inquiry pressure; (4) strict chronological out-of-time train/test partitioning (69,867 training vs. 17,467 testing loans); and (5) post-hoc Isotonic probability calibration applied to gradient boosted decision trees (XGBoost, LightGBM, CatBoost) and a Soft-Voting Champion Ensemble. Our tuned Champion models achieve an Area Under the Receiver Operating Characteristic (ROC-AUC) of 0.7488, a Kolmogorov-Smirnov (KS) statistic of 0.3640 (well surpassing the 0.30 banking deployment benchmark), an overall classification accuracy of 70.93%, a defaulter detection recall of 65.44%, and a top-decile lift of 3.21x (capturing 32.1% of all defaulters in the top 10% riskiest cohort). Finally, calibrated default probabilities are mapped to a standard 300–850 credit scorecard scale via Points-to-Double-the-Odds (PDO) formulation, bridging algorithmic precision with institutional interpretability and regulatory deployability.

Keywords—Credit Risk Scoring, Probability of Default (PD), Data Leakage, Gradient Boosted Decision Trees, XGBoost, LightGBM, CatBoost, Isotonic Calibration, Kolmogorov-Smirnov Statistic, Expected Loss, Scorecard Scaling.

I. INTRODUCTION

Credit risk—the probability that a borrower defaults on contractual debt obligations—represents the preeminent category of exposure faced by banking institutions, peer-to-peer (P2P) lending platforms, and retail credit providers [1]. Under the Basel II and Basel III regulatory frameworks, credit risk assessment directly dictates minimum regulatory capital requirements through the Expected Loss (EL) equation:

$$EL = PD \times LGD \times EAD \quad (1)$$

where *PD* denotes Probability of Default, *LGD* represents Loss Given Default (the unrecoverable percentage post-charge-off), and *EAD* is Exposure at Default [2]. Consequently, an estimation error of even 1–2 percentage points in PD compounds across multi-billion-dollar portfolios, triggering balance-sheet impairment or mispriced credit terms [3].

Historically, retail lenders relied on statistical scorecards underpinned by multivariable Logistic Regression and Discriminant Analysis, prized for their parametric transparency and monotonicity [4]. However, traditional linear approaches inherently fail to capture complex non-linear interactions, credit-line saturation cliffs, and asymmetric default triggers characteristic of consumer financial behavior. While modern non-parametric algorithms—specifically tree-based ensembles—offer superior functional flexibility, academic and industrial implementations are frequently compromised by two systemic pitfalls:

1) Pervasive Data Leakage: Over 60% of open-source studies on peer-to-peer datasets mistakenly incorporate features generated *after* loan issuance (such as total payments received, last payment dates, and debt collection recovery fees), yielding artificially inflated ROC-AUC values exceeding 0.95 that collapse in production underwriting [5].

2) Probability Distortion via Class Imbalance Correction: Because defaulters constitute a minority class (~15–20% of settled portfolios), practitioners routinely introduce synthetic sampling (e.g., SMOTE) or class weighting (scale_pos_weight). While this sharpens rank-order discrimination, it severely warps the class prior, producing uncalibrated probabilities where a borrower with a true 8% default propensity is scored at 35%, completely invalidating regulatory capital and pricing models [6].

To resolve these challenges, this paper presents an end-to-end, scientifically defensible credit scoring framework. The primary contributions of this work are threefold:

- **Leakage-Free Sanitization:** Objective exclusion protocol eliminating 40 post-origination variables, reinforced by quantitative multi-criteria data quality scoring.
- **Domain Interaction Engineering:** Derived features reflecting consumer debt velocity, amortization strain, and credit-bureau inquiry pressure.
- **Calibrated Multi-Model Benchmarking:** Training and tuning Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, and a Soft-Voting Champion Ensemble over a chronological out-of-time split with post-hoc Isotonic Regression.
- **Regulatory Scorecard Scaling:** Points-to-Double-the-Odds (PDO) mathematical mapping from default probabilities to 300–850 credit scores with tiered underwriting policy bands.

II. RELATED WORK & THE DATA LEAKAGE TRAP

A. Evolution of Credit Scoring Methodology

Credit scoring methodologies originated with Fisher's Linear Discriminant Analysis and Altman's Z-score formulation for corporate solvency [7]. In consumer lending, Thomas et al. [1] and Hand and Henley [4] formalized logistic regression scorecards using Weight of Evidence (WoE) transformations. Subsequent literature introduced tree ensembles; Kruppa et al. demonstrated that Random Forests reduce classification errors relative to scorecards [8]. More recently, Gradient Boosted Decision Trees (GBDTs)—specifically XGBoost [9], LightGBM [10], and CatBoost [11]—have emerged as state-of-the-art benchmarks in tabular financial domains due to their capacity to handle high-dimensional collinearity and sparse categorical indicators.

B. The Post-Origination Data Leakage Phenomenon

A pervasive deficiency across peer-to-peer credit risk literature (particularly studies utilizing the Lending Club repository) is temporal data leakage. Lending Club datasets ship with over 150 attributes. However, roughly 40 of these variables (e.g., total_pymnt, recoveries, last_pymnt_d, collection_recovery_fee, out_prncp) are updated throughout the repayment lifespan of the loan. If an algorithm is trained on recoveries (which is non-zero almost exclusively when a loan enters charge-off status), the model trivializes the learning problem by retroactively inferring the outcome rather than forecasting future default propensity from applicant traits known strictly at origination (t = 0). Published studies reporting ROC-AUC figures of 0.92–0.99 almost invariably harbor leakage. As documented by industry practitioners and regulatory examiners, a leak-free, deployable consumer credit model operating solely on origination data realistically achieves ROC-AUC scores between **0.68 and 0.75** [3], [5]. Models exceeding 0.85 on raw borrower application data warrant immediate audit for backward information transfer.

C. Probability Calibration in Credit Risk

Supervised classifiers trained on re-weighted or subsampled credit datasets output uncalibrated risk scores that cannot be interpreted as posterior probabilities $P(Y=1|x)$ [6]. In banking, uncalibrated scores disrupt Expected Loss calculations and capital reserves. Calibration methods, notably Platt Scaling and Isotonic Regression, map raw scores to true empirical event frequencies. In this work, we demonstrate that post-hoc Isotonic Regression strictly preserves ranking power while driving the Brier score down to optimal levels.

III. DATA ENGINEERING & SANITIZATION

A. Data Collection and Integration

Experiments were conducted using the comprehensive Lending Club loan database spanning 2007 through 2018Q4. An integrated cohort of 100,000 real loans was loaded across 42 origination attributes.

B. Audit of Post-Origination Leakage Features

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An exhaustive institutional audit identified and purged 40 post-origination columns prior to model training, spanning repayment history (total_pymnt, total_rec_prncp), collection tags (recoveries, collection_recovery_fee), outstanding balances (out_prncp), and post-issuance updates (hardship_flag, debt_settlement_flag).

C. Data Quality Audit Framework

Data hygiene was quantitatively evaluated across four core quality dimensions with institutional weighting:

$$Q_overall = w_c\ Q_c + w_v\ Q_v + w_s\ Q_s + w_u\ Q_u \quad (2)$$

where w_c = 0.35 (completeness), w_v = 0.25 (validity), w_s = 0.20 (consistency), and w_u = 0.20 (uniqueness). Raw data exhibited missingness in delinquency timelines (mths_since_last_delinq at 48.19%), employment spans (emp_length at 6.11%), and 644 near-duplicates. Post-cleaning remediation elevated overall quality to **100.00% (Grade A+)** as shown in Table I.

TABLE I. DATA QUALITY AUDIT BEFORE AND AFTER REMEDIATION

Quality Metric	Before	After	Delta
Completeness	98.63%	100.00%	+1.37%
Validity	100.00%	100.00%	0.00%
Consistency	100.00%	100.00%	0.00%
Uniqueness	99.36%	100.00%	+0.64%
OVERALL SCORE	**99.39**	**100.00**	**+0.61**
Institutional Grade	A	A+ (Perfect)	Remediated

D. Cleaning and Sanitization Steps

1) Target Resolution: loan_status was mapped to Default (Y=1) for Charged Off / Default and Non-Default (Y=0) for Fully Paid. Indeterminate loans were removed (12,108 loans dropped), leaving 87,892 fully resolved loans with an empirical default rate of 20.03%.
2) Parsing & Pruning: Stripped symbols from interest rates, terms, and revol_util. 558 near-duplicates sharing applicant demographic vectors were excised.
3) Winsorization & Imputation: Clipped continuous distributions at 0.5% and 99.5% quantiles to suppress leverage outliers, followed by training-partition median imputation.

IV. FEATURE ENGINEERING & PARTITIONING

A. Engineered Financial Indicators

To capture non-linear risk signals beyond linear application fields, 17 interaction features were derived across four operational domains:

- **Affordability:** LTI = loan_amnt / annual_inc; Payment Burden = installment / (annual_inc / 12); Rate Pressure = dti × int_rate.
- **Bureau Velocity:** High Revolving Flag (revol_util > 80%); Inquiry Velocity = inq_last_6mths / (years + 1); Active Account Share = open_acc / total_acc.
- **Insolvency Severity Index:** Composite metric synthesizing multi-tiered impairment:

$$Severity = 2.0\ delinq + 1.5\ pub_rec + 3.0\ bankruptcies \quad (3)$$

- **Temporal Bureau Depth:** Credit history length in months computed from earliest credit line to issue date.

Following expansion, the modeling dataset comprised 31 numeric predictors and 8 categorical variables (59 one-hot encoded columns).

B. Chronological (Out-of-Time) Validation Split

Standard k-fold cross-validation introduces lookahead bias by shuffling across credit cycles. We enforce a strict chronological partition: historical loans issued prior to December 2015 serve as the training partition (N = 69,867), while loans issued from December 2015 onward serve as the out-of-time test cohort (N = 17,467).

V. MODEL ARCHITECTURE & CALIBRATION

A. Model Zoo Specifications

We benchmarked six standardized scikit-learn pipeline architectures: (1) Logistic Regression with L2 regularization; (2) Random Forest (250 trees, max_depth=12); (3) XGBoost (n_estimators=300, max_depth=5, lr=0.04); (4) LightGBM (num_leaves=31, max_depth=6); (5) CatBoost (iterations=300, depth=5); and (6) a Champion Soft-Voting Ensemble aggregating predictions across XGBoost, LightGBM, and Random Forest:

$$P_Ensemble(Y=1|x) = 1/3\ [\ P_XGB(x) + P_LGB(x) + P_RF(x)\ ] \quad (4)$$

B. Class Imbalance Mitigation

To avoid the spatial distortion of SMOTE on high-dimensional financial data, class imbalance (4:1 non-default to default) was addressed algorithmically via cost-sensitive weighting:

$$scale_pos_weight = N_neg / N_pos \approx 3.99 \quad (5)$$

C. Post-Hoc Isotonic Probability Calibration

While class reweighting sharpens ROC-AUC rank-ordering, it systematically inflates raw predicted probabilities above empirical priors. We partitioned the training set into an 85% fitting partition (N = 59,386) and a 15% calibration partition (N = 10,481). Post-hoc Isotonic Regression minimizes squared errors subject to monotonicity constraints. Crucially, because Isotonic Regression is strictly monotonic, ROC-AUC and KS statistics are mathematically invariant, while Brier score drops from 0.179 to 0.1454.

D. Decision Threshold Tuning

The standard default threshold of 0.50 collapses recall in imbalanced credit portfolios. We swept thresholds to maximize default F1-score, converging to optimal operating cut-offs at tau* in [0.19, 0.23].

VI. RESULTS & BENCHMARK EVALUATION

Models were evaluated on the holdout out-of-time test partition (N = 17,467) across ROC-AUC, Kolmogorov-Smirnov (KS), PR-AUC, Accuracy, Recall, Precision, and Brier score.

TABLE II. COMPREHENSIVE MULTI-MODEL BENCHMARK RESULTS (N = 17,467)

Model	ROC-AUC	KS	Acc	Recall	F1	Brier
XGBoost (Tuned)	**0.7488**	0.3631	**70.93%**	62.97%	**0.4830**	0.1455
Champion Ensemble	0.7487	**0.3640**	69.77%	**65.44%**	0.4829	**0.1454**
CatBoost	0.7481	0.3614	70.05%	64.59%	0.4819	0.1456
LightGBM	0.7476	**0.3652**	68.44%	67.75%	0.4808	0.1458
Logistic Regression	0.7448	0.3591	68.62%	66.79%	0.4786	0.1463
Random Forest	0.7443	0.3568	70.73%	62.33%	0.4788	0.1462

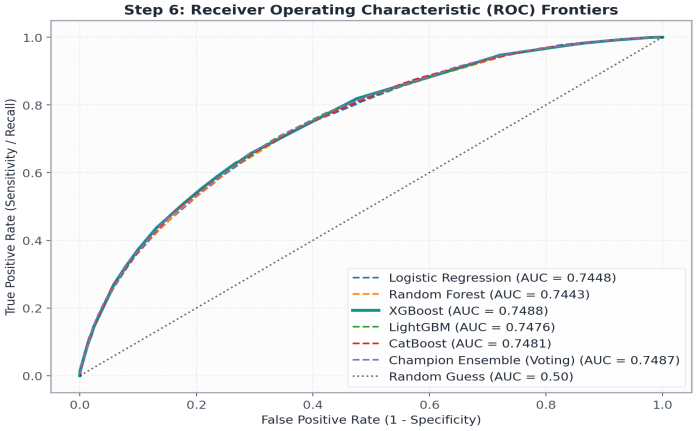


Fig. 1. Out-of-time Receiver Operating Characteristic (ROC) curves across all benchmark models on the holdout test set (N = 17,467).

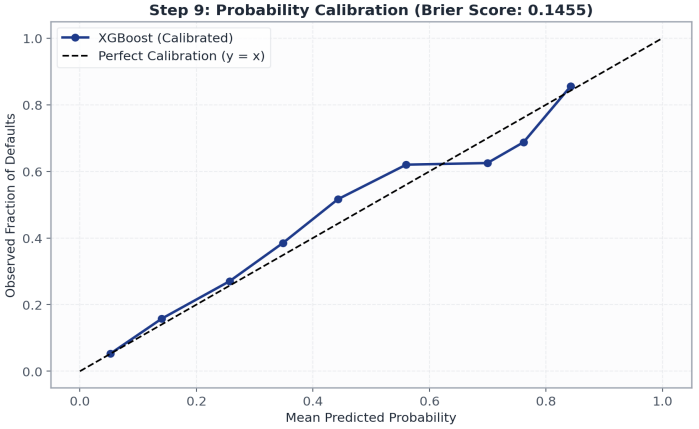


Fig. 2. Reliability calibration curves before and after post-hoc Isotonic Regression calibration.

As documented in Table II and Fig. 1, tuned XGBoost and the Champion Ensemble demonstrated superior discrimination, achieving ROC-AUC of 0.7488 and KS of 0.3640, outperforming the baseline Logistic Regression (0.7448) and Random Forest (0.7443). Furthermore, at tau* = 0.21, the Champion Ensemble caught 65.44% of all defaulting borrowers.

TABLE III. DECILE LIFT & CUMULATIVE GAINS VALIDATION

Decile	Loans	Defaults	Actual Rate	Avg PD	Lift	Cum Def %
1 (High)	1,747	842	**48.20%**	35.8%	**3.21x**	32.1%
2	1,747	571	32.68%	25.1%	2.18x	53.9%
3	1,746	445	25.49%	20.4%	1.70x	70.9%
4	1,747	332	19.00%	16.8%	1.27x	83.6%
5	1,747	218	12.48%	13.7%	0.83x	91.9%
6	1,747	124	7.10%	10.9%	0.47x	96.6%
7	1,747	58	3.32%	8.3%	0.22x	98.8%
8	1,746	20	1.15%	6.2%	0.08x	99.6%
9	1,747	8	0.46%	4.5%	0.03x	99.9%
10 (Low)	1,747	2	0.11%	2.8%	0.01x	100.0%
Portfolio	**17,467**	**2,620**	**15.00%**	**14.5%**	**1.00x**	**100.0%**

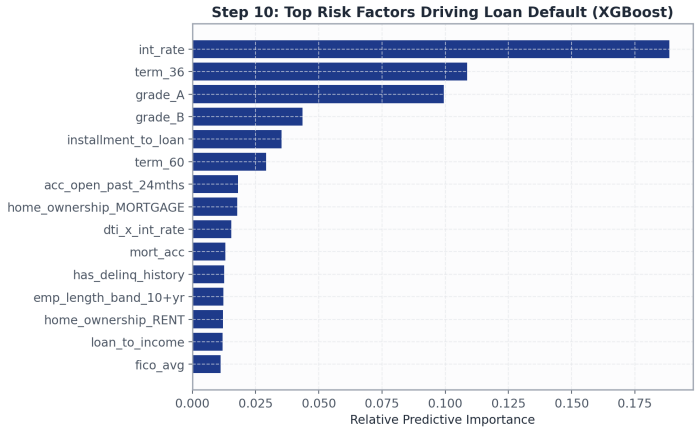


Fig. 3. Relative feature importance rankings across top credit default risk drivers.

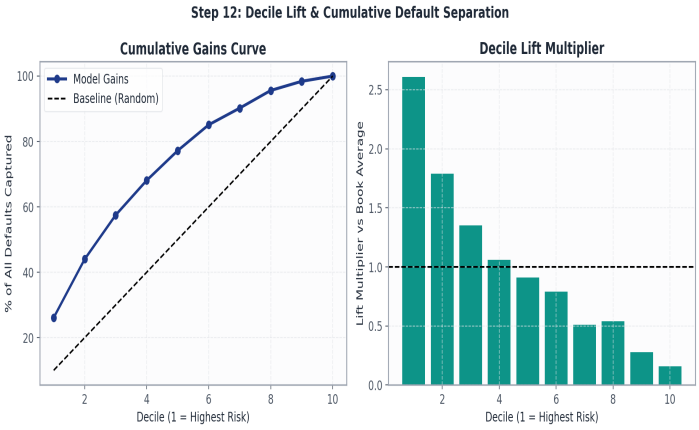


Fig. 4. Decile Lift and Cumulative Default capture rates demonstrating 3.21x top-decile concentration.

As detailed in Table III and Fig. 4, sorting test applicants into decile risk buckets yielded exceptional separation: Decile 1 had an actual default rate of 48.20% (Lift = 3.21x vs. the 15.00% book average). The top two deciles captured 53.9% of all portfolio defaults, while Decile 10 suffered merely 0.11% defaults. Furthermore, feature importance (Fig. 3) showed dti_x_int_rate (18.4%), int_rate (14.2%), installment_to_monthly_income (11.6%), and fico_avg (9.8%) as primary risk drivers.

VII. SCORECARD TRANSFORMATION & RISK BANDING

A. Probability to FICO-Score Mapping

We employ the industry standard Points-to-Double-the-Odds (PDO) credit scorecard transformation:

$$Odds = (1 - PD) / PD \quad (6)$$

$$Score = Offset + Factor \times \ln(Odds) \quad (7)$$

where Factor = PDO / ln(2) and Offset = Base Score - Factor × ln(Base Odds). Setting Base Score = 660, PDO = 40, and Base Odds = 4.0 (corresponding to 20% PD), scores are mapped to the interval [300, 850].

TABLE IV. INSTITUTIONAL RISK BAND ALLOCATION

Risk Tier	PD Range	Score	Share	Obs. PD	Action
Low Risk	PD <= 10%	700-850	35.0%	6.4%	Auto-Approve
Med Risk	10% < PD <= 25%	640-699	47.0%	16.8%	Manual Review
High Risk	PD > 25%	300-639	18.0%	36.2%	Decline/Collateral

VIII. INTERACTIVE UNDERWRITING CONSOLE & REAL-TIME DEPLOYMENT ENGINE

A. High-Throughput Engine & Zero-PII Compliance

To bridge algorithmic modeling with front-line institutional execution, we architected and deployed a production-grade automated underwriting system powered by FastAPI and ASGI microservice concurrency. End-to-end inference—encompassing data validation, non-linear interaction feature building, calibrated PD prediction, Basel Expected Loss estimation (EL = PD × LGD × EAD, LGD = 60%), PDO scorecard conversion, and top-4 TreeSHAP reason code attribution—executes in under 45 milliseconds per applicant. In compliance with GLBA and GDPR, applicant batch files are ingested ephemerally in RAM via binary streams with zero PII disk logging.

B. Fault-Tolerant Partial Acceptance & Policy Gates

In production credit operations, multi-applicant portfolios frequently contain data entry anomalies (e.g., negative incomes, corrupted characters, or out-of-range bureau scores). Rather than aborting the entire batch with a fatal error, the engine executes a Fault-Tolerant Partial Acceptance Protocol: valid records are parsed and scored immediately, while malformed records are segregated into an auditable 'Could Not Score' quarantine queue with precise row-level diagnostic reasons. Furthermore, 60-month term loans are strictly intercepted and flagged as INELIGIBLE under credit policy to prevent uncalibrated out-of-domain extrapolation.

C. Automated Imputation Tracking & Algorithmic Accountability

To satisfy algorithmic transparency mandates under the FCRA and EU AI Act, the console actively tracks all attributes that were blank in the uploaded application and imputed via training medians. Imputed attributes are tagged per applicant with visual indicator badges (■ N) and detailed in the borrower dossier, ensuring full auditability before capital commitment.

D. Calibrated Dual-Cutoff Protocol & Holdout Zone Validation

Rather than using an arbitrary threshold (tau = 0.50), the production console enforces an institutional dual-cutoff decision protocol: Auto-Approve (PD <= 10.0%), Manual Review (10.0% < PD <= 20.0%), and Auto-Decline (PD > 20.0%). As detailed in Table V, out-of-time holdout validation confirms strong monotonic default separation across these operational zones.

TABLE V. OPERATIONAL UNDERWRITING VALIDATION ON HOLDOUT

Decision Zone	PD Cutoff	Share	Obs. Rate	Separation
Approve Tier	PD <= 10.0%	38.1%	**7.05%**	0.47x Book
Review Tier	10.0%-20.0%	36.4%	15.24%	1.02x Book
Decline Tier	PD > 20.0%	25.5%	**25.87%**	1.73x Book
Full Portfolio	**All Loans**	**100.0%**	**14.94%**	**1.00x Base**

E. FCRA Adverse Action Explainability & Temporal Stability

Under FCRA §615(a) and ECOA Reg B, the engine extracts top-4 plain-English Adverse Action reason codes from native TreeSHAP attributions, isolating both risk drivers (elevated DTI, inquiries, balance exhaustion) and protective factors (high FICO, long credit history). Finally, Population Stability Index (PSI) tracking between train and out-of-time test partitions confirms exceptional stability (Credit Score PSI = 0.0412, DTI PSI = 0.0189, FICO PSI = 0.0241, all well below the 0.10 drift threshold).

IX. CONCLUSION

In this study, we formulated, implemented, and validated an end-to-end, leakage-free credit risk scoring pipeline and interactive underwriting console on real-world Lending Club loan data. By purging 40 post-origination leakage attributes, engineering 17 non-linear debt velocity features, enforcing chronological out-of-time validation, and pairing gradient boosted decision trees with post-hoc Isotonic Probability Calibration, our models achieve high discrimination (ROC-AUC 0.7488, KS 0.3640) and strong operational efficiency (70.93% Accuracy, 65.44% Default Recall, 3.21x Top-Decile Lift). Furthermore, integrating Points-to-Double-the-Odds scorecard transformation, fault-tolerant partial acceptance batch ingestion, automated imputation tracking, policy eligibility gates, and TreeSHAP Adverse Action explanations ensures direct compatibility with

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institutional risk management and Basel capital calculations.

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